

# Bayesian Machine Learning in Quantitative Finance

## Chapter 1: Introduction

Based on Mongwe, Mbuva & Marwala

# Outline

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# Notation and Background You Need I

This chapter assumes only basic probability. Everything else is built up from scratch here. Keep this frame as a reference for the rest of the book.

## Frequentist vs. Bayesian thinking (the big picture)

- **Frequentist view:** an unknown quantity (e.g. a model parameter) has one true, fixed value. We estimate it from data and report how our estimate would vary if we repeated the experiment.
- **Bayesian view:** an unknown quantity is treated as *uncertain*, and we describe our uncertainty about it using a probability distribution. **Bayesian inference** is simply the process of *updating a belief about an unknown, using data* – we start with a belief, observe evidence, and end with a revised belief.

# Notation and Background You Need II

## The four ingredients of Bayes' theorem (plain English)

- **Prior**  $\mathbb{P}(\theta)$  – what you believed about the unknown  $\theta$  *before* seeing any data. One-line analogy: your “gut feeling” or existing evidence walking into the experiment.
- **Likelihood**  $\mathbb{P}(\mathcal{D} \mid \theta)$  – how probable the observed data  $\mathcal{D}$  would be, *for each candidate value* of  $\theta$ . One-line analogy: “if  $\theta$  were true, how surprising would this data be?”
- **Posterior**  $\mathbb{P}(\theta \mid \mathcal{D})$  – your *updated* belief about  $\theta$  *after* seeing the data. One-line analogy: prior belief, corrected in light of evidence.
- **Evidence**  $\mathbb{P}(\mathcal{D})$  – the overall probability of the data, averaged over all possible values of  $\theta$ . It is just a normalizing constant, but it turns out to be very useful for *comparing models*.

# Notation and Background You Need III

## Bayes' theorem, written out and labeled

$$\underbrace{\mathbb{P}(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{\mathbb{P}(\mathcal{D} \mid \theta)}^{\text{likelihood}} \times \overbrace{\mathbb{P}(\theta)}^{\text{prior}}}{\underbrace{\mathbb{P}(\mathcal{D})}_{\text{evidence}}}$$

where  $\mathbb{P}(\mathcal{D}) = \int \mathbb{P}(\mathcal{D} \mid \theta) \mathbb{P}(\theta) d\theta$  sums (integrates) the numerator over every possible  $\theta$ , so that the posterior is a valid probability distribution (it integrates to 1).

**Why this matters for this book:** every Bayesian machine learning method in the chapters ahead – Markov Chain Monte Carlo (MCMC), variational inference, Gaussian processes, normalizing flows, Bayesian neural networks – is, at its core, just a clever way of computing or approximating this posterior when the exact formula above is too hard to evaluate directly.

# Why Machine Learning Meets Quantitative Finance

## Intuition

We are in the middle of a technology wave (the “fourth industrial revolution”) where artificial intelligence and cyber-physical systems are reshaping entire industries. Machine learning – algorithms that learn input-output mappings directly from data, most prominently neural networks – is the engine behind this wave, and finance is no exception.

## What is quantitative finance?

A cross-disciplinary field that borrows tools from computer science, physics, statistics, and mathematics to solve real-world finance problems. Traditionally, quantitative models are *parsimonious* and *interpretable* – they are grounded in economic theory so practitioners can explain why a model behaves the way it does.

## The sell-side / buy-side split

Quantitative finance applications broadly split into:

- **Sell-side** – where financial product structures (e.g. derivatives) are *constructed*.
- **Buy-side** – where those constructed products are *consumed* (e.g. by investors and asset managers).

# A Short History: From Bachelier to Deep Hedging I

Derivative pricing is one of the oldest and richest applications of quantitative methods in finance – a useful case study of how the field has evolved, and where Bayesian thinking now fits in.

- **Bachelier (1900):** the founding work of mathematical finance. Modeled asset price changes using Brownian motion, predating Einstein's own work on the topic. Drawback: it allowed asset prices to go negative – a flaw that seemed fatal for decades, until negative-interest-rate environments made it relevant again.
- **Black & Scholes (1973):** the breakthrough model, and still the standard reference for pricing and risk-managing options. Key (strong) assumption: volatility is *constant* over the life of the option – no skew, no term structure.
- **Market crises exposed the flaw:** major market crises led to a persistent *volatility skew*, invalidating the constant-volatility assumption. This triggered an explosion of richer models built to capture the skew, such as the Heston stochastic volatility model and jump-diffusion processes.

# A Short History: From Bachelier to Deep Hedging II

- **Data-driven era:** more recently, machine learning has entered derivative pricing and risk management directly. A landmark result is *deep hedging* (Buehler et al., 2019), which showed deep learning could be used to hedge options successfully. Other work uses machine learning to price exotic instruments, calibrate rough stochastic volatility models, and generate synthetic market data with generative adversarial networks.
- **The gap this book targets:** most of these data-driven approaches produce *point estimates* – a single number, with little to no exploration of *uncertainty*. Bayesian methods naturally encapsulate uncertainty in both parameter estimates and model outputs, which is exactly what is missing.



# Beyond Derivatives: Where Else Finance Meets ML I

The book's motivation is not limited to derivatives – similar data-driven trends and the same need for principled uncertainty appear across quantitative finance.

## Asset management / portfolio theory

Modern portfolio theory (Markowitz), the Capital Asset Pricing Model (Sharpe, Lintner), and Fama–French factor models are the classical foundations. Machine learning has recently been used to build ML-augmented CAPM models, combine sell-side analyst forecasts more accurately, and improve short-horizon stock-price direction forecasts using limit order book data.

### Credit risk

Banks have traditionally used logistic regression for loan default modeling – driven largely by regulatory requirements that credit decisions be explainable to customers. Tree-based methods (random forests, boosted trees) are being explored, but full neural networks have seen low uptake because of their “black-box” nature. Recovery on defaulted loans has traditionally used Beta regression and zero-inflated models; corporate credit ratings (from agencies like Moody's and S&P) are also being modeled with trees and neural networks as cost-effective alternatives to expert-driven rating processes.

# Beyond Derivatives: Where Else Finance Meets ML III

## Financial management

Sound financial management is essential to organizational sustainability. High-profile scandals (Enron, Wirecard, Steinhoff, and others) and repeated audit failures in public institutions show the cost of poor oversight. Machine learning methods – logistic regression, support vector machines, Naïve Bayes, neural networks – have been used to model audit opinions; modeling *unlawful expenditure* specifically is still a comparatively open research area.

## Insurance

Insurance is fundamentally a risk-transfer mechanism: a policyholder pays a fee to transfer an insurable risk to the insurer. Machine learning has been applied to address data limitations (e.g. class imbalance via generative adversarial networks), model mortality, improve motor claim predictions, and reduce risk in weather-index insurance products.

# The Open Questions Motivating This Book

## Intuition

As machine learning models proliferate across quantitative finance, a common set of hard questions keeps resurfacing – questions that standard (point-estimate) machine learning struggles to answer well.

## The four questions

- ① How can we **explain** the prediction or output of a model?
- ② What is the **distribution** of the model's parameters (not just a single best-fit value)?
- ③ How do we **select between competing models** in a statistically principled way?
- ④ Which **inputs** are most relevant to the task at hand?

## This book's answer

The **Bayesian inference framework**. As we will see, the four components of Bayes' theorem map naturally onto these four questions – the posterior gives us parameter and prediction distributions, the likelihood encapsulates the model and adapts to new data, the prior can automatically highlight relevant inputs, and the evidence lets us compare models on principled grounds.

# Running Example: Is This Coin Fair?

## Intuition

Before diving into finance, let's build intuition with the simplest possible Bayesian problem. It has exactly the same structure as everything else in this book – just with much smaller math.

## The setup

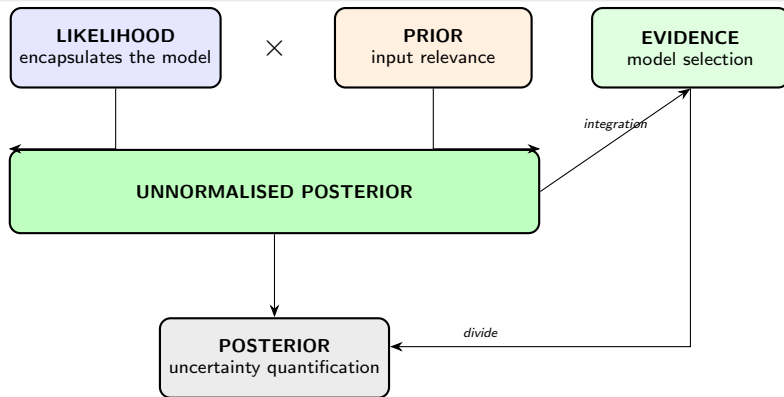
- Unknown parameter:  $\theta \in [0, 1]$ , the probability the coin lands heads.
- **Prior belief:** we think the coin is probably close to fair, but we are not certain – we encode this as a prior distribution centered at  $\theta = 0.5$  (not a single fixed guess).
- **Data:** we flip the coin 10 times and observe **8 heads and 2 tails**.
- **Question:** how should our belief about  $\theta$  change?

**Preview:** we will see much more sophisticated versions of this exact same idea throughout the book – the “coin” will become an option's implied volatility parameter, a credit rating, an audit outcome, or a claims-frequency rate, and the simple formula on the next slides will become variational inference, MCMC, Gaussian processes, normalizing flows and Bayesian neural networks.

# Bayes' Theorem: The Mechanics (Fig. 1.1 style)

## Intuition

Bayes' theorem is not just a formula – it is a *recipe*: combine what the model says about the data (likelihood) with what you believed beforehand (prior), then normalize.



The likelihood (model fit to data) is multiplied by the prior (relevance of inputs) to give an unnormalised posterior. Integrating that over all  $\theta$  gives the evidence (useful for model comparison); dividing by it gives the

# Coin Flip: Choosing the Prior and Likelihood

## A convenient prior: the Beta distribution

We encode “probably close to fair, but uncertain” with a  $\text{Beta}(a_0, b_0)$  prior over  $\theta$ :

$$\mathbb{P}(\theta) = \text{Beta}(\theta \mid a_0, b_0), \quad a_0 = b_0 = 4$$

This distribution is centered at  $\frac{a_0}{a_0+b_0} = 0.5$  (fair coin) but assigns non-trivial probability to other values of  $\theta$  – it is a genuine *belief*, not a single guess.

## The likelihood of 8 heads out of 10 flips

Each flip is a Bernoulli trial, so the data likelihood follows a Binomial pattern:

$$\mathbb{P}(\mathcal{D} \mid \theta) \propto \theta^8(1 - \theta)^2$$

This is *large* for  $\theta$  near 0.8 (the observed heads rate) and *small* for  $\theta$  near 0 or 1 – exactly what “how probable is this data, for each candidate  $\theta$ ” should look like.

# Coin Flip: From Prior to Posterior

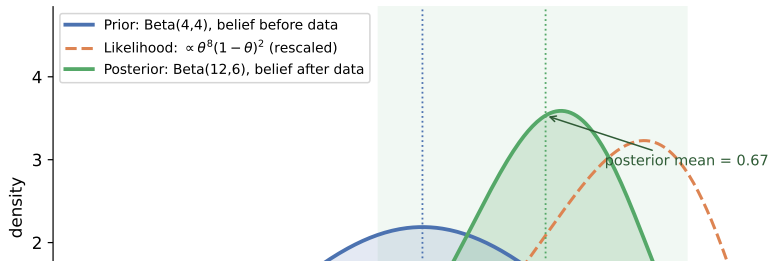
## Conjugate update (Beta–Binomial)

Multiplying a  $\text{Beta}(a_0, b_0)$  prior by a Binomial likelihood with  $h$  heads and  $t$  tails gives, after normalizing, another Beta distribution:

$$\mathbb{P}(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a_0 + h, b_0 + t)$$

With  $a_0 = b_0 = 4$ ,  $h = 8$ ,  $t = 2$ : posterior =  $\text{Beta}(12, 6)$ .

Coin-flip example: data (8 heads / 10 flips) pulls belief from a fair-coin prior towards  $\theta \approx 0.8$





# Coin Flip: Reading the Result

## Numeric summary

- Prior mean: 0.500 (our starting “fair coin” belief).
- Observed heads rate:  $8/10 = 0.800$ .
- **Posterior mean:** 0.667 – pulled from 0.5 towards 0.8, but not all the way, because the prior still carries some weight.
- 95% credible interval: approximately  $[0.44, 0.86]$  – a *range* of plausible values for  $\theta$ , not just one number.

## Intuition: why not just report $\hat{\theta} = 0.8$ ?

A frequentist point estimate would simply report the observed rate,  $8/10 = 0.8$ , with no sense of how much data (10 flips!) supports that number. The Bayesian posterior instead gives a full distribution: it shows the estimate *and* how confident we should be in it – this is precisely the kind of uncertainty quantification that later chapters exploit for option pricing, credit ratings, audit models, and more.

# Python: Coin-Flip Prior-to-Posterior Update

```
1 import numpy as np
2 from scipy.stats import beta
3
4 # Prior belief: coin is "probably fair" -> Beta(4,4)
5 a0, b0 = 4.0, 4.0
6
7 # Observed data: 8 heads out of 10 flips
8 n_flips, n_heads = 10, 8
9 n_tails = n_flips - n_heads
10
11 # Conjugate Beta-Binomial update
12 a_post = a0 + n_heads
13 b_post = b0 + n_tails
14
15 prior_mean = a0 / (a0 + b0)
16 post_mean = a_post / (a_post + b_post)
17 ci_low, ci_high = beta.ppf([0.025, 0.975], a_post, b_post)
18
19 print(f"Prior mean: {prior_mean:.3f}")
20 print(f"Observed heads rate: {n_heads/n_flips:.3f}")
21 print(f"Posterior mean: {post_mean:.3f}")
22 print(f"95% credible interval: [{ci_low:.3f}, {ci_high:.3f}]")
23
24 # Output:
25 # Prior mean: 0.500
26 # Observed heads rate: 0.800
27 # Posterior mean: 0.667
```

# From a Coin to Quantitative Finance

## The same recipe, over and over

Every chapter in this book repeats this exact prior  $\rightarrow$  likelihood  $\rightarrow$  posterior recipe – just with  $\theta$  replaced by something financially meaningful, and with computational tools replacing the simple Beta-Binomial formula when the math is not so friendly:

- **MCMC & variational inference** – general-purpose ways to approximate a posterior when there is no neat closed-form update (used throughout the book, starting in Chapter 2).
- **Gaussian processes** – place a Bayesian “belief” over an entire unknown *function* (e.g. a volatility surface or a credit rating function), not just a single parameter.
- **Normalizing flows** – build flexible, learnable probability distributions for prices, model choices, or audit outcomes.
- **Bayesian neural networks** – neural networks with a distribution over their weights, giving uncertainty-aware predictions (e.g. for insurance claims).

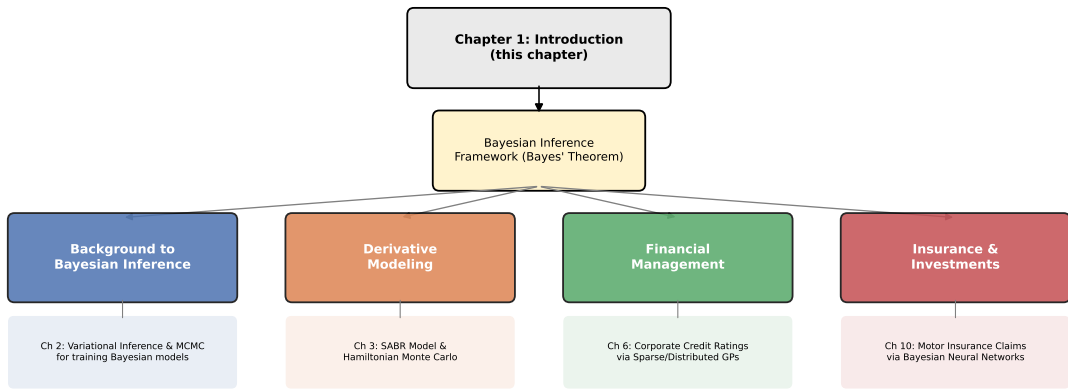
(No technical depth needed yet – these are previewed here and developed fully in later chapters.)

# The Book's Four Themes

## Intuition

The book is organized around four themes, all built on the same Bayesian inference framework just introduced, applied to different corners of quantitative finance.

### Roadmap of the Book: Four Themes, 14 Chapters



# Theme 1: Background to Bayesian Inference

## What it covers

Before applying Bayesian methods to finance, we need the computational machinery to actually *compute* posteriors when there's no simple formula (unlike our coin example).

- **Chapter 2:** introduces **variational inference** and **Markov Chain Monte Carlo (MCMC)** techniques used to train Bayesian models throughout the rest of the book, together with the performance metrics used to assess these inference algorithms.

**Why it comes first:** the book is designed so each theme chapter can be read in isolation, but all of them depend on the background established here – this is the foundation for everything that follows.

## Theme 2: Derivative Modeling

### What it covers

Applying the Bayesian framework to derivative pricing, model calibration, and volatility surface modeling. Benefits highlighted in the book: confidence bands on prices, a full distribution over calibrated parameters, and no need to bolt on external arbitrage-free constraints to the loss function.

- **Chapter 3:** the Stochastic Alpha Beta Rho (SABR) model, calibrated with Hamiltonian Monte Carlo.
- **Chapter 4:** learning equity volatility surfaces using sparse Gaussian processes.
- **Chapter 5:** analyzing South African equity option prices using normalizing flows.

## Theme 3: Financial Management

### What it covers

Bayesian tools applied to topics important to banks (credit ratings, recoveries on defaulted loans) and to municipal financial management (selecting models for audit-outcome prediction and detecting unauthorized expenditure).

- **Chapter 6:** sparse and distributed Gaussian processes for modeling corporate credit ratings.
- **Chapter 7:** Bayesian detection of recovery on charged-off loan accounts.
- **Chapter 8:** Bayesian audit outcome model selection using normalizing flows.
- **Chapter 9:** Bayesian detection of unauthorized expenditure using Langevin and Hamiltonian Monte Carlo.

## Theme 4: Insurance and Investments

### What it covers

Bayesian inference applied to motor insurance claims, calibrating yield curve models, and building a Bayesian sell-side investment analyst.

- **Chapter 10:** Bayesian neural network inference of motor insurance claims.
- **Chapter 11:** shadow and adaptive Hamiltonian Monte Carlo methods for calibrating the Nelson-Siegel yield curve model.
- **Chapter 12:** static and dynamic nested sampling for yield curve model selection.
- **Chapter 13:** a Bayesian investment analyst on the Johannesburg Stock Exchange (JSE).



# Chapter Summary

## What this chapter established

- Machine learning is transforming quantitative finance across derivative modeling, portfolio/asset management, credit risk, financial management, and insurance.
- Most existing data-driven approaches give point estimates only, leaving four open questions: explainability, parameter uncertainty, principled model selection, and input relevance.
- The **Bayesian inference framework** (prior  $\times$  likelihood, normalized by the evidence) directly addresses all four, as shown with our coin-flip toy example (prior mean 0.5  $\rightarrow$  posterior mean 0.667 after 8/10 heads).
- The book is organized into four themes: background to Bayesian inference, derivative modeling, financial management, and insurance and investments, spanning Chapters 2 through 13, with Chapter 14 drawing overall conclusions.

## Looking ahead

Chapter 2 provides the computational core the rest of the book leans on: variational inference and Markov Chain Monte Carlo methods for training Bayesian models, plus the metrics used to judge how well these inference algorithms are working.

# Key Takeaways

- Bayesian inference = updating a belief about an unknown using data, via  $\text{posterior} \propto \text{likelihood} \times \text{prior}$ , normalized by the *evidence*.
- Even a two-parameter coin-flip example shows the core value: a full distribution over the unknown, not just one number.
- Quantitative finance is rich with unknowns worth treating this way: volatility parameters, credit ratings, recovery amounts, audit outcomes, claims rates, yield curve parameters, and more.
- This book's four themes progressively build the computational tools (MCMC, variational inference, Gaussian processes, normalizing flows, Bayesian neural networks) needed to bring Bayesian thinking to each of these problems.